

Overview of Migraine Causes, Symptoms, and Treatments and Potential New Treatment Targets

Introduction

Migraine is a complex and incapacitating neurological illness that affects millions around the world. Understanding the complex interaction of migraine origins, symptoms, and treatment options is critical for successful management and better patient outcomes. This literature review provides a thorough examination of the various elements of migraines, focusing on the underlying mechanisms that contribute to their onset and progression. We hope to shed insight on the various etiological routes that contribute to migraine development by investigating genetic, environmental, and pathophysiological factors. Furthermore, a thorough analysis of the wide spectrum of symptoms experienced during migraine attacks provides useful insights into the condition's diagnostic and clinical presentations. This review also dives into the existing therapeutic options, such as acute and preventive drugs, lifestyle changes, and alternative therapies, that have shown efficacy in migraine management. Furthermore, we investigate potential novel treatment targets and developing medicines that have the potential to improve therapeutic outcomes. This review aims to provide a comprehensive understanding of the causes, symptoms, current treatment modalities, and potential new avenues for migraine management by synthesizing and analyzing the existing literature, thereby contributing to the advancement of knowledge in this field and informing future research and clinical practice.

Causes Of Migraine

Migraines are a complex neurological illness characterized by acute and recurring headaches that are frequently accompanied by other symptoms such as nausea, light and sound sensitivity, and vision problems. While the specific origins of migraines are unknown, researchers and healthcare experts have found a number of factors that can contribute to the development and frequency of these incapacitating headaches [2, 4]. Migraines are thought to be caused by a mix of genetic, environmental, and physiological variables that interact to initiate the neurological cascade that causes the typical symptom [9, 17]. Understanding the underlying causes of migraines is critical for effective management and therapy because it allows healthcare providers to design individualized techniques to alleviate symptoms and enhance the quality of life for those who suffer from this illness.

Migraine Risk Factors

Genetics

The contribution of genetics to migraine is very complex. 38 different loci have been associated with the occurrence of migraine in individuals through various population studies [11, 17]. The high number and functional difference between these loci suggests that migraine is likely the result

of interactions between multiple genes and epigenetic factors. Those that have relatives with migraine have a higher risk of also having migraine. Recent studies estimate the heritability of migraine to be around 42% [12].

Migraine development and susceptibility are heavily influenced by genetics. Numerous studies have found a strong familial component, with individuals with affected family members having a higher prevalence of migraines. It is estimated that heredity accounts for 50-60% of the overall risk of migraine development [11, 12]. Researchers discovered many genetic variations linked to an increased risk of migraines, with some genes exhibiting a greater link than others. These genes participate in a variety of biological functions, including as neurotransmitter modulation, ion channel function, and vascular homeostasis. Variations in serotonin signaling genes, such as the 5-HT1B and 5-HT1D receptors, have been linked to migraine vulnerability. Furthermore, mutations in ion channel genes, such as CACNA1A and SCN1A, have been associated to certain migraine subtypes, such as familial hemiplegic migraine¹ [12, 13, 14].

Understanding the genetic basis of migraines not only offers light on the condition's fundamental causes, but also holds hope for the future development of targeted medicines and tailored approaches to migraine care.

Gender and Sex Hormones

Gender and sex hormones have been demonstrated to have a major impact on migraines. Women get migraines at a higher rate than men, according to studies, suggesting that sex hormones may play a role in the pathophysiology of migraines [20, 21, 28]. Women's migraines have a strong correlation to the change of estrogen levels during the menstrual cycle [20, 21, 22]. An increased risk of migraines has been linked to the reduction in estrogen levels during the premenstrual phase and the days immediately following menstruation [21, 22]. Additionally, the occurrence and intensity of migraines can be affected by hormonal changes such as pregnancy, menopause, and the use of hormonal contraceptives [19, 23]. It is thought that changes in estrogen levels may have an impact on the sensitivity of the trigeminal nerve, which is involved in the pathways that cause migraine pain, even if the precise processes underpinning these hormonal impacts on migraines are not entirely understood [28]. The established modulation of neurotransmitters and vascular function by estrogen may also contribute to the onset of migraines [20, 22]. It is crucial to know

¹ Hemiplegic migraine is a rare and distinct kind of migraine that causes transient paralysis or weakness on one side of the body [12, 14]. This crippling condition is distinguished by intense headaches that are accompanied by neurological symptoms such as vision problems, trouble speaking, and sensory alterations. Hemiplegic migraines are further subdivided into familial hemiplegic migraine (FHM), which has a genetic component, and sporadic hemiplegic migraine (SHM), which does not [12, 15]. The precise origin of hemiplegic migraines is unknown, however it is thought to entail abnormalities in brain electrical activity and neurotransmitter imbalances. Hemiplegic migraines can be severely incapacitating and necessitate rapid medical attention to rule out more serious illnesses such as stroke. Preventive strategies, lifestyle changes, and acute treatment options tailored to the individual's specific needs are commonly used to manage hemiplegic migraines [15, 16].

how gender, sex hormones, and migraines interact in order to customize treatment plans and offer people with this crippling ailment practical management techniques.

Migraine Triggers

Metabolism and Diet

An important part of the intricate network of elements that cause migraines is metabolism. Numerous metabolic changes in the body frequently accompany migraine headaches, which can either trigger or exacerbate these incapacitating episodes [17, 25]. Neurotransmitter levels being altered is one important way that metabolism affects migraines [14, 24]. Neurotransmitters including serotonin, dopamine, and norepinephrine, which have all been linked to the onset of migraines, are produced and broken down by metabolism [24]. The equilibrium of these neurotransmitters can be upset by changes in metabolism, which can start a migraine attack. Additionally, metabolic processes affect blood vessel regulation, and problems with these mechanisms may be a factor in vascular abnormalities associated with migraines [17, 26]. For instance, changes in the metabolism of nitric oxide, a powerful vasodilator, can cause aberrant blood vessel dilation and constriction, which causes the typical throbbing pain felt during migraines. Additionally, metabolic abnormalities, such as variations in glucose levels, can alter both the brain's susceptibility to pain signals and the availability of energy [17, 24]. The modulation of neurotransmitter levels, vascular control, and energy metabolism by metabolism has a multidimensional impact on migraines, emphasizing the need of addressing metabolic aspects in the treatment and prevention of migraines [24, 25]. Furthermore, diet and migraines have a bidirectional relationship rather than a simply causational one. Metabolism shifts and diet affect an individual's migraine severity, frequency, and symptoms. These, in return, affect the individual's dietary choices, patterns, and content [25].

Environmental changes

Environmental changes can have a significant impact on migraines, often acting as triggers or exacerbating factors for individuals prone to these debilitating headaches. Various environmental factors, such as changes in weather patterns, exposure to certain scents or chemicals, and variations in sleep habits, have been identified as potential migraine triggers [9, 18]. Weather variations, particularly abrupt temperature fluctuations, changes in barometric pressure, and high humidity levels, have been known to cause migraines in certain people [9, 10, 27]. Strong scents, such as perfumes, cigarettes, or certain chemicals, can also serve as triggers [10]. Sleep deprivation, especially changes in sleep patterns or irregular sleep schedules, has been linked to an increased risk of migraines [18]. Furthermore, environmental factors such as bright or flickering lights, loud noises, and excessive stress or emotional stimulus can all contribute to migraine onset [10, 18]. While the precise mechanisms underlying the effects of these environmental changes on migraines are undetermined, it appears that they can disrupt the delicate balance of neurotransmitters, alter vascular tone, or increase overall neuronal excitability, resulting in the onset of a migraine episode

[9, 10, 17]. Recognizing and addressing these environmental factors can be an important element of migraine prevention and management.

Migraine Pathophysiology

Phases of a Migraine Attack

Premonitory Phase

The premonitory phase, also known as the prodrome phase, is an important stage that precedes the actual onset of a migraine attack. This phase, which may begin hours or days before the headache develops, is marked by a variety of symptoms that can differ from person to person. Fatigue, mood swings, food cravings or aversions, increased urination, stiff neck, and cognitive difficulties are typical signs of the premonitory period [2, 4]. Additionally, some people may develop more modest neurological symptoms such as changes in sensory perception, difficulties concentrating, and visual problems [8, 17]. Individual differences can affect the length and severity of the premonitory period, with some people reporting modest symptoms and others experiencing more dramatic and prolonged manifestations [7]. Understanding the premonitory phase is critical because it allows for early intervention and preventive actions to reduce the severity and duration of the later headache phase.

PET and fMRI studies on both spontaneous and triggered migraine attacks have revealed intriguing findings about changes in hypothalamic activity preceding headache onset. Specific premonitory migraine symptoms, such as polyuria (increased urine production), mood swings, and hunger changes, have been linked to these alterations in hypothalamic activity [17]. Light sensitivity and nausea, both of which can be linked to the activation of the brainstem, have been found to be correlated with the increased activity seen in the occipital cortex during the premonitory phase [4, 5, 6]. These neuroimaging methods shed light on the interactions between different brain regions and the presentation of migraine symptoms, offering insightful information into the dynamic neurological processes taking place before a migraine attack. Additional investigation in this field may increase our understanding of the underlying causes of migraines and pave the way to more specialized and efficient treatments.

Electrophysiological investigations, focusing on thalamic and thalamo-cortical activity, have provided important insights into the intricate neurological processes causing migraines. During migraine attacks, studies have shown aberrant thalamic activity, including changes in neuronal firing patterns and disturbed synchronization with cortical regions [17]. These variations in thalamic and thalamo-cortical activity have been associated with abnormal sensory processing, which results in migraine symptoms such as sensitivity to light, sound, and touch [5, 6]. Understanding the atypical electrical patterns in the thalamus and thalamo-cortical circuits shows promise as a possible migraine treatment target. It may be possible to alleviate sensory disturbances and lessen the frequency and intensity of migraine attacks by regulating or restoring the normal functioning of certain brain networks.

Resting-state MRI studies have revealed changes in the interactions between several brain regions and the overlapping circuits related with sensory processing, pain processing, anxiety, and mood, both before and during a migraine attack. These data support the hypothesis that migraine has a major impact on the brain's general performance. The changes in resting-state MRI scans show that migraine's effects extend beyond specific regions and circuits, implying a widespread impact on brain activity and connectivity [17, 46].

Aura Phase

Approximately one third of migraine attacks include an aura phase which is recurrent attacks of sensory and other CNS symptoms [1]. It typically starts around an hour before the onset of the headache and lasts for anywhere from a few minutes up to an hour. The most prominent symptoms are visual disturbances (such as seeing flashing lights, blind spots, or zigzag lines), speech/language difficulties, and motor disturbances [1, 2, 17]. Although the exact mechanisms are not fully understood, the aura phase is thought to be caused by changes in electrical activity and blood flow in the brain.

The start of an aura phase is associated with Cortical Spreading Depression (CSD). Animal studies support that CSD activates trigeminal nociception which can trigger migraine attacks. CSD is a slowly progressing depolarization of the neuronal and glial cell membranes followed by the inhibition of cortical activity for approximately 30 minutes. The propagation of CSD is hypothesized to be a result of either K⁺ or glutamate interstitial diffusion or mediated by gap junctions between glial cells or neurons [2, 17]. However, clinical symptoms of aura do not precede the majority of migraine attacks; aura may develop after the headache phase has already begun, and individuals may have aura but not the subsequent headache [1, 3].

Headache Phase

The headache phase, which is widely accepted to be related to the trigeminovascular pathway which conveys nociceptive information from the meninges to the central areas of the brain and the cortex [2, 3, 4]. Because this pathway is responsible for the processing of sensory information, it is correlated with the common migraine symptoms such as photophobia, phonophobia, cognitive dysfunction, osmophobia, and allodynia [2, 4]. These symptoms can also be risk factors for developing chronic migraine [1]. Individuals often feel strong, pulsating headaches during this period, which may last anywhere from a few hours to multiple days [2, 17]. The discomfort typically limited to one side of the head, but it may affect both sides. Migraine headaches are frequently accompanied by a variety of distressing symptoms, such as light and sound sensitivity, dizziness, impaired vision, nausea and vomiting [17]. Migraine headaches can vary widely in length and severity from person to person, making each attack a distinct experience [14, 17].

Symptoms Before and During a Migraine Attack

Migraine symptoms can be extremely debilitating, impacting millions of people throughout the world. Migraines, which are characterized by strong and frequent headaches, may show a

complicated variety of symptoms that can impede daily life. Migraines appear in a variety of ways, each unique to the individual, ranging from pulsing throbbing feelings and high sensitivity to light and sound to nausea, vomiting, and even visual problems [1, 2, 17]. Understanding the wide spectrum of migraine symptoms is critical for properly detecting and controlling this neurological illness. Furthermore, recognizing the unique pattern of symptoms during migraine attacks can aid in differentiating migraines from other types of headaches and contribute to more accurate diagnoses and personalized treatment plans.

Current and Recently Developed Migraine Treatment Methods

Acute Treatment Methods

Three strategies of medication treatment

The stratified approach to migraine treatment, widely utilized in clinical practice, is predicated on the severity of the headache. This method involves adjusting the therapy regimen to the severity of the headaches reported by people [30, 37, 38]. Nonsteroidal anti-inflammatory drugs (NSAIDs) are often used to treat moderate headaches, whereas stronger treatments, such as triptans, are usually reserved for severe headaches [37]. According to existing studies, this stratified treatment strategy has a higher efficacy in obtaining pain relief within a two-hour timeframe. However, it is worth noting that this technique has been linked to an increased incidence of adverse effects [38].

The "step care across attacks" approach is a migraine treatment strategy that prioritizes drug selection based on criteria such as cost and potential side effects. In this approach, the initial drug choice is guided by price considerations as well as the possibility of adverse effects. If the initial treatment fails to provide significant relief, the next step is to switch to a more potent medication alternative [30, 37]. This method tries to optimize resource use and eliminate unnecessary risks while gradually increasing treatment efficacy [38].

The "step care within attacks" approach is a treatment strategy employed in the management of migraines that involves the initial use of simple analgesics and subsequent escalation to more potent medications within the same migraine attack if the initial treatment proves ineffective [30, 37]. This approach recognizes the dynamic nature of migraines, acknowledging that the severity and duration of symptoms can vary within an individual attack [38]. By initially utilizing simple analgesics, such as nonsteroidal anti-inflammatory drugs (NSAIDs), the aim is to address milder symptoms and minimize potential side effects. However, if the response to the initial treatment is suboptimal, a step-wise progression to more potent medications, such as triptans or ergots, is implemented to achieve effective symptom control [37]. Notably, this approach has demonstrated efficacy in patients who present with irregular symptom patterns and experience prolonged migraine episodes [30, 38].

With all treatment strategies, acute migraine patients must be warned about the danger of medication overuse and the possible conversion to chronic migraine [31, 37].

First-line treatments

The treatment of mild to moderate migraines commonly involves the administration of acetaminophen and oral nonsteroidal anti-inflammatory drugs (NSAIDs). Acetaminophen, a common analgesic, is effective in relieving the pain associated with mild to moderate migraines. Similarly, oral NSAIDs have been shown to be effective in relieving migraine-related discomfort in clinical trials [30, 35]. In fact, placebo-controlled trials have consistently demonstrated that NSAIDs give better pain relief than acetaminophen. However, it is crucial to recognize that NSAIDs do have certain downsides. Their use may have side effects such as gastric irritation or antiplatelet effects, which should be carefully considered, especially in people who have a history of gastrointestinal problems or bleeding disorders [35]. The combination of acetaminophen, aspirin, and caffeine have emerged as an effective first-line therapy choice for mild to moderate acute migraines. This medication therapy has been found to successfully treat migraine pain while minimizing the potential side effects associated with individual NSAIDs [39].

When dealing with moderate to severe migraines, the best first-line medication is triptans, which act as selective 5-hydroxytryptamine (5-HT) receptor agonists [29, 30]. Triptans work by inhibiting inflammation and restricting blood arteries in the brain by targeting certain serotonin receptors. The timing of triptan administration is critical for obtaining maximum efficacy, as they are most effective when employed early in a migraine episode [35]. However, due to the genetic variances underlying migraine pathophysiology, not all patients may experience a favorable reaction to every triptan. Genetic factors can influence the metabolism, receptor affinity, and overall therapeutic response to triptan medications, thereby necessitating personalized treatment strategies [30, 35].

Second-line treatments

When first-line migraine therapies, such as triptans, are ineffective or contraindicated, second-line migraine therapy options are available [30, 35]. Dihydroergotamine is one such alternative that has been shown to be useful in reducing migraine symptoms. However, when compared to triptans, dihydroergotamine has lower receptor selectivity, resulting in a higher prevalence of side effects [30, 41]. These side effects, while treatable, call for caution while administering this medicine. Opioids, such as butorphanol, codeine, tramadol, and meperidine, are another second-line therapeutic option that should be addressed with caution [30, 40]. While opioids can give pain relief, they are highly addictive and should be used with caution and under constant supervision. Antiemetics, which are dopamine antagonists, are also often given parenterally and have moderate evidence supporting their usefulness in migraine management [30]. However, long-term or excessive use of antiemetics can result in extrapyramidal side effects, which are infrequent with

intermittent usage [42]. Promethazine and Ketorolac are other commonly employed second-line treatments that are injected intramuscularly to relieve migraine symptoms [30, 40, 43].

Preventive Treatment Methods

Onabotulinumtoxin-A treatment (aka botox), which was first licensed for aesthetic purposes, has shown success in treating chronic migraines, which are defined as headaches on 15 or more days per month for at least 4 hours. Botox reduces the frequency and severity of migraine attacks by suppressing the release of particular neurotransmitters involved in pain signaling. It was shown to be effective with chronic migraine patients and resulted in reversion to episodic migraine [31, 33]. Botox is applied every 12 weeks to 31 standardized areas on the head and neck through injection (with an optional 40 units applied with a “follow the pain” method) [31]. Botox can cause brief adverse effects like localized pain or muscular weakness at the injection site, but these usually go away on their own. It is also important to recognize that Botox is not appropriate for everyone, and people with certain medical conditions or who are on certain drugs may not be good candidates for this procedure [33].

Interventions such as occipital nerve blocks are sometimes used to treat migraines [31]. These procedures entail the injection of local anesthetic and/or corticosteroids around the occipital nerves, which are located at the back of the head. The occipital nerves are responsible for sending pain signals from the back of the head to the brain. These nerves may become sensitive or irritated in some migraine patients, contributing to headache symptoms. Occipital nerve blocks provide comfort by momentarily numbing or relaxing these nerves, halting pain impulses [44, 45]. Occipital nerve blocks have been shown to be useful in lowering migraine frequency, duration, and intensity in clinical research [31]. They may also be effective for people who have not responded well to previous preventive treatments or who are experiencing adverse effects from other medication treatments. Occipital nerve blocks are frequently used along with other pharmacological and non-pharmacological migraine treatment options [31, 44, 45].

Other nondrug therapies such as acupuncture or behavioral interventions like diet changes have also shown to be helpful with preventive management with migraine [17, 31, 47].

Recently Developed Treatment Methods

Significant advances in migraine therapy have occurred in recent years, leading to the development of novel therapeutic options that hold promise for patients suffering from this severe neurological disorder. These new therapies are intended to address several elements of migraines, such as the underlying pathophysiology, symptom relief, and future episode prevention.

5-HT as a target

5-HT is a nonvascular serotonergic non-triptan receptor which suggests that it doesn't play a role in the vasoconstriction of microcirculation of the brain [29, 36]. Lasmiditan is highly selective for

5-HT receptors and lacks vasoconstrictive properties, which makes it a suitable option for patients with CVD (who can't use triptans) [36]. By the activation of the 5-HT receptors, *trigeminal neuron activity* is blocked and the acute migraine pathway is inhibited [29].

CGRP as a target

CGRP receptor antagonists control thalamic activity as a result of trigeminal nociceptive stimulation [36]. One way triptans work to relieve pain is by decreasing CGRP levels in the migraine pathway [29, 31]. There are other CGRP antagonists (small molecules like gepants, ubrogepant and rimegepant; or larger ones like mAbs). These CGRP antagonists can be used to inhibit vasodilation and neurogenic inflammation by blocking the release of CGRP throughout the migraine pathway and work as a preventive treatment method [29, 36].

Erenumab, fremanezumab, and galcanezumab (3 CGRP mAbs) were approved by the FDA in 2018 for migraine treatment in adults. They are all administered monthly subcutaneously, and the most common adverse effect was irritation at the injection site [36].

Botox (Onabotulinumtoxin-A)

BoNT-A was initially hypothesized to relieve pain through muscle relaxation and following hypotension. Different mechanisms such as inhibiting the exocytosis of neurochemicals and proteins of the motor and sensory systems, reducing exocytosis of proinflammatory cells, neurotransmitters, and excitatory neuropeptides of the nervous system (substance P, CGRP, glutamate), and inhibiting soluble N-ethylmaleimide-sensitive factor attachment protein receptors (SNARE) were proposed as to how BoNT-A functions [31, 33]. Recent studies, however, have shown that BoNT-A might actually be affecting the CNS. Several studies support BoNT-A efficacy on decreasing migraine frequency and severity with the only common side effects being neck pain and muscle weakness [33].

Conclusion

This literature review provides a comprehensive overview of migraine causes, symptoms, and existing treatment options, while also highlighting potential new treatment targets for this complex neurological condition. Migraines continue to be a serious global health burden, affecting millions of people and severely reducing their quality of life. We have obtained vital insights into the multidimensional nature of migraine pathophysiology by investigating the underlying mechanisms, which include genetic, environmental, and neurovascular variables. Furthermore, a detailed evaluation of the wide range of symptoms reported during migraine attacks has highlighted the necessity for holistic and customized treatment approaches. While established treatment approaches, including as acute and preventive drugs, lifestyle changes, and alternative therapies, have been shown to be effective, there is a growing corpus of research focusing on identifying novel therapy targets. By advancing our understanding of migraine causes, symptoms, and treatment, this literature review contributes to the current discussion about migraine management, as well as laying the groundwork for future research and therapeutic breakthroughs

in the topic. Finally, the incorporation of new therapy targets, in conjunction with comprehensive care techniques, provides hope for migraine sufferers, with the goal of reducing the burden of this condition and improving overall patient outcomes and quality of life.

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